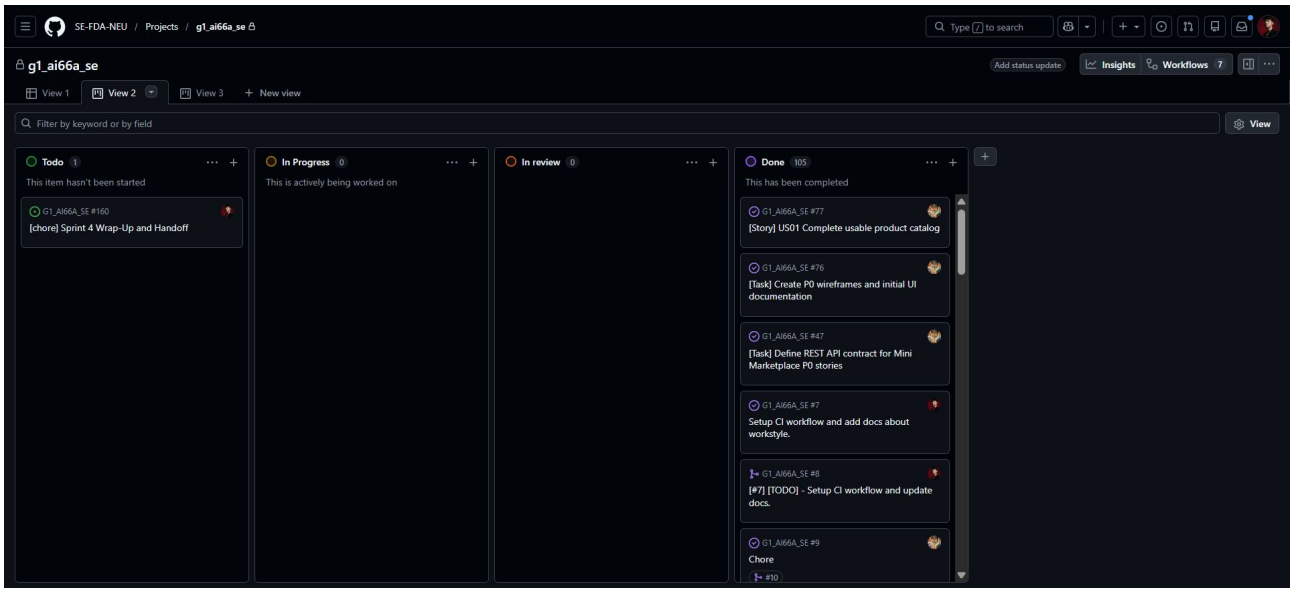
Team01_M4 — Milestone 4

Team: Team 01 — Mini Marketplace (G1_AI66A_SE)
Topic: Mini Marketplace — Milestone 4: automated testing
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Repository: https://github.com/SE-FDA-NEU/G1_AI66A_SE
Project board: https://github.com/orgs/SE-FDA-NEU/projects/6
Setup guide: https://github.com/SE-FDA-NEU/G1_AI66A_SE/blob/main/docs/SETUP.md

Submitted by: Nguyen Quang Minh

Page 2 — Project board screenshot



Page 3 — Output of `python -m pytest -v` (full suite, middle omitted)

```
============================= test session starts ==============================
platform linux -- Python 3.12.3, pytest-9.1.1, pluggy-1.6.0 -- /home/minhquang/G1_AI66A_SE/.venv/bin/python
rootdir: /home/minhquang/G1_AI66A_SE
configfile: pyproject.toml
testpaths: tests
plugins: anyio-4.15.1, cov-7.1.0
collecting ... collected 1690 items
tests/test_addresses.py::test_crud_defaults_and_persistence PASSED       [  0%]
tests/test_addresses.py::test_ownership_auth_and_stale_version_do_not_clear_defaults PASSED [  0%]
tests/test_addresses.py::test_address_validation[patch0] PASSED          [  0%]
tests/test_addresses.py::test_address_validation[patch1] PASSED          [  0%]
tests/test_addresses.py::test_address_validation[patch2] PASSED          [  0%]
tests/test_addresses.py::test_address_validation[patch3] PASSED          [  0%]
tests/test_addresses.py::test_address_validation[patch4] PASSED          [  0%]
tests/test_addresses.py::test_address_validation[patch5] PASSED          [  0%]
tests/test_addresses.py::test_address_validation[patch6] PASSED          [  0%]
tests/test_addresses.py::test_address_validation[patch7] PASSED          [  0%]
tests/test_addresses.py::test_address_validation[patch8] PASSED          [  0%]
tests/test_addresses.py::test_address_validation[patch9] PASSED          [  0%]
tests/test_addresses.py::test_address_validation[patch10] PASSED         [  0%]
tests/test_addresses.py::test_address_validation[patch11] PASSED         [  0%]
tests/test_addresses.py::test_address_validation[patch12] PASSED         [  0%]
tests/test_addresses.py::test_address_validation[patch13] PASSED         [  0%]
tests/test_addresses.py::test_invalid_address_shapes[payload0] PASSED    [  1%]
tests/test_addresses.py::test_invalid_address_shapes[text] PASSED        [  1%]
tests/test_addresses.py::test_invalid_address_shapes[None] PASSED        [  1%]
tests/test_addresses.py::test_invalid_address_shapes[payload3] PASSED    [  1%]
tests/test_addresses.py::test_strict_address_versions[True] PASSED       [  1%]
tests/test_addresses.py::test_strict_address_versions[1] PASSED          [  1%]
tests/test_addresses.py::test_strict_address_versions[0] PASSED          [  1%]
tests/test_addresses.py::test_strict_address_versions[-1] PASSED         [  1%]
tests/test_addresses.py::test_strict_address_versions[1.0] PASSED        [  1%]
tests/test_addresses.py::test_constraint_no_id_reuse_and_init_upgrade PASSED [  1%]
tests/test_addresses.py::test_default_failure_rolls_back_clear PASSED    [  1%]
tests/test_addresses.py::test_concurrent_default_operations_keep_one_default PASSED [  1%]

...

ERROR tests/test_admin_portal_browser.py::test_category_crud_and_keyboard_modal_use_real_persistence - AssertionError: Locator expected t
============ 1689 passed, 4 warnings, 1 error in 1822.46s (0:30:22) ============
```

Testing (Milestone 4)

Run everything: `python -m pytest -v` (setup in SETUP.md). Run only the P0 test cases of section 2: `python -m pytest tests/test_m4_p0.py -v`. Per-task regression notes written earlier in the project are kept in testing-task-notes.md.

1. What we test, and why

Layer	Test	Why
Services (`src/services/`: cart, checkout, pricing, order states)	unit tests, plus service tests on a temporary SQLite file	Business rules live here (stock limits, idempotency, price maths, status transitions); a mistake here costs money or stock
Repository / database (`src/models`, `src/database.py`)	tests on a temporary SQLite file with foreign keys on	Checks constraints, rollback after failure and migrations
Routes (`src/api/`)	FastAPI `TestClient` against a temporary database (`tests/test_m4_p0.py`, `test_p0_flows.py`)	Status codes, authorization by role, and that a refused request writes nothing
Interface (`src/web/static/*.js`, HTML)	DOM scripts (`node tests/*_ui.mjs`) and browser tests in CI; not part of `tests/test_m4_p0.py`	Checked by users in Milestone 3; browser tests are slower and need Chrome, so they are not the M4 P0 cases

Risk: we tested checkout and cart most heavily (atomic stock deduction, idempotency key, stale stock, concurrent requests, rollback), because they move money and inventory and a partial write is hard to repair. Seller product limits and the order status state machine come next: they are rules with numbers or a table where an off-by-one or a wrong transition is easy to introduce. Catalog and detail pages are read-only, so they get fewer cases.

Expected values in section 2 come from the requirements (limits, prices, statuses), not from reading the production constants back.

2. Test cases for every P0 story

Kind: N = normal, E = error (a rule refuses the input), B = boundary.

ID	Story	Input	Expected result	Kind
TC-US01-01	US01 Catalog	one active product, search by its name	listed in `GET /api/v1/products`	N
TC-US01-02	US01 Catalog	product deactivated by its seller	not listed	E
TC-US01-03	US01 Catalog	19 products, page size 20	19 on page 1, total 19	B, below
TC-US01-04	US01 Catalog	20 products, page size 20	20 on page 1, total 20	B, on the limit
TC-US01-05	US01 Catalog	21 products, page size 20	20 on page 1, 1 on page 2	B, above
TC-US02-01	US02 Product detail	existing code, price 123 000	200, price 123 000	N
TC-US02-02	US02 Product detail	unknown code	404	E
TC-US02-03	US02 Product detail	stock 0	200, stock_quantity 0	B
TC-US02-04	US02 Product detail	stock 1	200, stock_quantity 1	B
TC-US03-01	US03 Cart	add 2 × 100 000 VND	201, cart total 200 000	N
TC-US03-02	US03 Cart	stock 10, add 9	201, cart holds 9	B, below
TC-US03-03	US03 Cart	stock 10, add 10	201, cart holds 10	B, on the limit
TC-US03-04	US03 Cart	stock 10, add 11	409/422, cart empty	B, above
TC-US03-05	US03 Cart	quantity 0	400, nothing saved	E, boundary
TC-US03-06	US03 Cart	quantity −1	400, nothing saved	E
TC-US04-01	US04 Checkout	cart 2 × 100 000, stock 5	201, total 200 000, stock 3	N
TC-US04-02	US04 Checkout	empty cart	4xx, no order saved	E
TC-US04-03	US04 Checkout	stock dropped 3 → 2 after adding 3	409, no order, stock stays 2	E
TC-US04-04	US04 Checkout	stock 2, quantity 2	201, stock 0	B, on the limit
TC-US04-05	US04 Checkout	same Idempotency-Key twice	201 then 200, one order	E (replay)

ID	Story	Input	Expected result	Kind
TC-US06-01	US06 Create product	seller, valid body	201, price 50 000	N
TC-US06-02	US06 Create product	buyer account	403	E
TC-US06-03	US06 Create product	price −1	refused	B
TC-US06-04	US06 Create product	price 0	refused	B
TC-US06-05	US06 Create product	price 1	201	B, smallest valid
TC-US06-06	US06 Create product	price 1 000 000 000	201	B, on the limit
TC-US06-07	US06 Create product	price 1 000 000 001	refused	B, above
TC-US06-08	US06 Create product	stock −1	refused	B
TC-US06-09	US06 Create product	stock 0	201	B
TC-US06-10	US06 Create product	stock 1 000 000	201	B, on the limit
TC-US06-11	US06 Create product	stock 1 000 001	refused	B, above
TC-US08-01	US08 Seller orders	order containing seller's product	seller lists exactly that order	N
TC-US08-02	US08 Seller orders	other seller lists / updates it	empty list; 403/404	E
TC-US08-03	US08 Seller orders	PENDING → CONFIRMED	200, CONFIRMED	N
TC-US08-04	US08 Seller orders	CONFIRMED → DELIVERED	400, status stays CONFIRMED	E
TC-US08-05	US08 Seller orders	7 pairs (PENDING→COMPLETED, CONFIRMED→SHIPPED, …)	decision table allowed / refused	E, table
TC-US08-06	US08 Seller orders	19 orders, limit 20	19 on page 1, total 19	B, below
TC-US08-07	US08 Seller orders	20 orders, limit 20	20 on page 1, total 20	B, on the limit
TC-US08-08	US08 Seller orders	21 orders, limit 20	20 on page 1, total 21	B, above

Note on US08: `docs/requirements.md` describes "Mark as Completed" from PENDING, but the state machine (`src/services/order_states.py`) requires PENDING → CONFIRMED/PROCESSING → SHIPPED → … → COMPLETED. TC-US08-05 follows the state machine; the team must reconcile the acceptance criterion with it.

3. Automated tests

All cases above are in tests/test_m4_p0.py; the ID is in the test name (`test_tc_us03_02_to_04_…`) and in the `# TC-…` comment of each parametrize row. Tests use a temporary SQLite file per test and the FastAPI `TestClient` (no server, no `marketplace.db`). Pure decision-table cases (TC-US08-05) call the service function directly.

```
python -m pytest -v                       # whole suite
python -m pytest tests/test_m4_p0.py -v    # the 45 M4 P0 cases
```

Output of the M4 file: docs/images/m4/pytest-m4.txt (45 passed). Existing suites (about 1 650 tests: checkout, cart service, pricing, rollback, concurrency, browser tests) continue to run in CI.

4. What the tests found

Bug: two checkout tests were silently never run. Ruff reported F811 (redefined function) in CI, which was allowed to fail. `tests/test_checkout.py` defined `test_checkout_persists_selected_payment_method_and_receipt` and `test_checkout_omitted_payment_method_uses_legacy_cod_compatibility` twice; Python kept the later, weaker copies, which dropped the `payment_status == "UNPAID"` and receipt assertions. Fix: the duplicate

copies were removed so the stronger tests run.

Rule broken on purpose (red output). In `src/services/cart.py` `validate_quantity`, we changed `quantity < 1` to `quantity < 0`, so quantity 0 is accepted. Result:

```
FAILED tests/test_m4_p0.py::test_tc_us03_05_06_non_positive_quantity_is_refused[0]
E   assert 409 == 400
1 failed, 1 passed
```

The code was restored (`git checkout`) and the test is green again. The test that guards this rule is TC-US03-05.